Custom Fence Panels

HackerRank

You are tasked with covering a fence of total length N using panels from a catalog. There are two standard panel types available in unlimited supply:

• Panels of length a, costing p each.

• Panels of length b, costing q each.

Additionally, you can order at most one custom panel of any positive integer length. Producing this custom panel costs a flat price of F, regardless of its chosen length.

The panels must be laid end-to-end to cover the fence of length N exactly. This means the sum of the lengths of all used panels must equal N. You cannot cut the standard panels, nor can they overhang the fence.

Find the minimum total cost to cover the fence exactly.

Input Format

The first line contains a integer, N, denoting the total length of the fence.

The second line contains a integer, a, denoting the length of the first standard panel type.

The third line contains a integer, b, denoting the length of the second standard panel type.

The fourth line contains a integer, p, denoting the cost of each panel of length a

The fifth line contains a integer, q, denoting the cost of each panel of length b

The next line contains a integer, F, denoting the flat cost of producing at most one custom panel

Constraints

1 <= N <= 10^9

1 <= a <= b - 1

a+1 <= b <= 10^9

1 <= p <= 10^9

1 <= q <= 10^9

1 <= F <= 10^9

Output Format

The output contains an integer value

Sample Input 0

```
10
3
4
10
15
100
```

Sample Output 0

```
35
```

Explanation 0

Using two standard panels of length 3 and one of length 4 exactly covers the fence of length 10 for a cost of 35, which is cheaper than paying the high custom panel fee of 100.

Sample Input 1

```
10
3
4
10
15
20
```

Sample Output 1

```
20
```

Explanation 1

Paying the flat fee of 20 for a single custom panel of length 10 is more cost-effective than using standard panels, which would cost 35.

Sample Input 2

```
5
3
4
10
10
100
```

Sample Output 2

```
100
```

Explanation 2

Since it is impossible to sum standard panels of lengths 3 and 4 to exactly 5, ordering a single custom panel of length 5 for 100 is the cheapest valid option.